

Question	Marking Guidance	Mark	Comments
04.1	1. Males have one allele; 2. Females need two recessive alleles OR Females must be homozygous recessive OR Females could have dominant and recessive alleles OR Females could be heterozygous/carriers;	2	1 Accept males only need one allele. 1 and 2. Ignore references to X and Y chromosomes. 1 and 2. Accept r as recessive allele and R as dominant allele. If no reference to allele, accept for one mark male needs one recessive gene whereas females need two recessive genes.

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04.2	1. Box 2. All females red-eyed, all males white-eyed.	1	Reject if more than one box with tick. Ignore crossed-out ticks.

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04.3	1. The (two) genes are linked OR Autosomal linkage;	3	1. Accept that the genes are on the same chromosome.

	2. No crossing over (occurs) OR (Linked) genes are close together; 3. No Gl and no gL (gametes produced) OR No Ggll and no ggLL (offspring produced) OR Only GL and gl (gametes produced);		1. Accept 'Alleles are linked' (accept symbols for alleles) but reject if context suggests alleles of the 'same gene'. 2. Accept crossing over less likely to occur. 1 2, and 3. Ignore reference to independent assortment.
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04.4	1. Correct answer of $8 \times 10^{10} = 3 \text{ marks};;$ 2. Correct answer not in standard form = 2 marks OR $1.6 \times 10^{13} = 2 \text{ mark}$ OR $1.6 \times 10^{11} = 2 \text{ mark}$ OR $6.4 \times 10^{11} = 2 \text{ mark}$ OR Shows 8×10^{10} in the working = 2 marks;; 3. $1.28 \times 10^{12} = 1 \text{ mark}$ OR $3.2 \times 10^{11} = 1 \text{ mark}$ OR $8 \times 10^{11} = 1 \text{ mark}$ OR $8 \times 10^9 = 1 \text{ mark}$ OR Shows 1.6×10^{11} in the working = 1 mark OR Shows 200^4 in the working = 1 mark;	3	If no other mark is credited accept for one mark working which shows multiplication by 200 for 4 generations. This could be shown in a variety of ways e.g. multiplied by 400 divided by 2 for 4 generations.

